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TITLE

- **Problem Statement ID – SIH26040**
- **Problem Statement Title- Smart Water Purification and Quality Monitoring System for Rural and Mining-Affected Areas.**
- **Theme- Clean & Green Technology**
- **PS Category- Hardware**
- **Team ID- SIH26-A0H-T206**
- **Team Name (Registered on portal): Pheonix**

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SMART WATER PURIFICATION

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Existing Challenge

Environmental Impact:
Soil Health Alteration, Water Scarcity, Impact on Aquatic & Terrestrial Biodiversity, Sedimentation & Erosion

Economic Impact:
Agricultural losses,
Reduced productivity,
Reduced Economic Growth,
Industrial losses

Health Impact:
Waterborne diseases,
Weak immunity,
Reduced well being of humans,
Heavy Metal Poisoning

Our Solution

Our solution is a **portable smart purification unit** that continuously evaluates water quality using multiple sensors. It identifies the specific quality issues and activates only the required purification stage instead of applying unnecessary treatment. Real-time readings and alerts help users monitor water safety, while its compact and energy-efficient design makes it practical for remote communities.

DETECT → ANALYZE → TREAT → PURIFY

Key Features

Automatic Treatment Adjustment

Targeted Purification — treats only what is required

Multi-Parameter Testing — pH, TDS, turbidity & temperature

Quality Status / Alerts & Post-Treatment Verification

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TECHNICAL APPROACH

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Initialize system components

- Initialize ESP32 + Sensors + Display
  ↓
- ESP32 → LCD/Display
  ↓
- ESP32 → Wi-Fi/ Bluetooth → Mobile Dashboard

Collect and read samples

- Collect water sample
  ↓
- Read all sensors: pH, TDS, Turbidity, Temperature

Analyze sensor data

- Merge & analyze sensor data
  ↓
- Are all parameters within limits?

pH ------- 7
Temperature ----22 C
Turbidity----0.3NTU

Treat water if necessary

- Identify parameters out of range
  ↓
- Select required treatment for pH, TDS, or Turbidity
  ↓
- Activate only required module

Verify water quality

- Re-test ALL parameters
  ↓
- Within limits? If yes, safe water; if no, repeat required treatment

FEASIBILITY AND VIABILITY

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Smart Water Purification&Quality Monitoring System

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1.ANALYSIS OF THE FEASIBILITY OF THE IDEA

Technical Feasibility
Uses proven,readily available components and IoT technology.
Technically achievable with current resources

Economic Feasibility
Low-cost components and cost effective

Operational Feasibility
Measures well a project.Fits within day-to-day cultures,operations and human resources

Scalability Feasibility
It can be scaled for homes,institutions and community use

Environmental Feasibility
Promotes water,reduces wastage and sustainable usage

2.POTENTIAL CHALLENGES AND RISKS

Sensor Accuracy
May vary due to environmental conditions

Maintenance Requirements
Sensors,filters and pumps need regular cleaning and replacement.

Purification Efficiency
Differnet contaminants require different filtration methods and maintenance

Power & Connectivity Issues
Poer failure and poor internet connectivity may affect monitoring and alerts

False Reading/System Failure
Sensor malfunction or system errors may result in incorrect decisions

3.STRATEGIES FOR OVERCOMING THESE CHALLENGES

Regular Calibration
Calibrate sensors periodically and perform routine accuracy checks

Rodundancy & Validation
Use multiple sensors and threshold validation to ensure reliable reading

Optimized filtration & Maintenance
Use multi-stage filtration and provide maintenance alerts for timely replacements

Backup & Offline Mode
Provide manual override,local storage and offline operation during power or internet failures.

System Diagnostics
Implement self-diagnostics and error alerts to detect and resolve issues early

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IMPACT AND BENEFITS

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KEY BENEFITS

1.Early contamination detection:
Sensors instantly spots harmful bacteria, heavy metal spikes.

2.Early Alerts:
Notifications can identify abnormal water quality.

3.Lower Operating Burden:
Efficient treatment and reduced dependence on manual support.

4.Consistent Water Quality:
Systems adjust filtration levels automatically based on detected impurities in the raw water supply.

EXPECTED IMPACTS

1.Public Health and Medical Impacts:
Stops waterborne epidemics by alerting communities , Chronic Disease reduction, Child Mortality Drops.

2.Economic and Financial Impacts:
Increased Work productivity, lower infrastructure maintenance costs.

3.Environmental and Resource Impacts:
Zero overhead water waste, Reduced plastic footprints, Aquifer health mapping.

4.Governance and Policy impacts:
Total data transparency, Evidence based funding,Instant emergency response.

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RESEARCH AND REFERENCES

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